

TABLE II
DEMOGRAPHY OF RURAL INDIA

Village population band	Number of villages	Population	Average population per village	Average households per village*	Percentage of total rural population
Less than 200	82,151	8,179,551	100	11	0.98
200-499	114,732	39,685,424	345	73	4.76
500-999	141,800	103,321,330	724	154	12.39
1000-1999	139,164	197,536,058	1410	300	23.69
2000-4999	96,428	288,773,884	2974	633	34.64
5000-9999	18,652	123,877,458	6,630	1411	14.86
10,000 and above	4681	72,375,147	16,026	3410	8.68
Total	597,608	833,748,852	1251	266	100

* Average household size in rural India = 4.6 (national sample survey)

Source: Census of India 2011 and national sample survey 68th Round (July 2009 - June 2010)

participate in radio communication. From the cognitive radio perspective, spectrum spaces are categorised into white space (totally unoccupied), gray space (partially occupied) and black space (fully occupied).

Accordingly, three techniques are used: interweaving, underlaying and overlaying. Interweaving refers to sensing the spectrum to identify unused bands and utilising the same for transmission. Underlaying refers to transmitting information at low power levels so as to not cause interference to licensed primary users. Overlaying refers to using part of the transmitting power to enhance occupied signal levels, thus increasing the probability of signal detection by licensed users. Simultaneously, information is transmitted at low power levels with other cognitive users.

Need for broadband connectivity in rural India

As per the census of India 2011, around 69 per cent of the people live in rural areas in approximately 600,000 villages (Table I). Villages can have wide population ranges with over 50 per cent having population between 1000 and 5000 (Table II).

The following observations give broad indications of the social-economic state of rural India:

1. Only 32 per cent of grade III students in government-run schools can read grade I textbooks

2. 31 per cent of rural population travels more than 30km to seek healthcare

3. 61 per cent of rural population remains unbanked

4. 41 per cent of farmers require vital agricultural information on a daily basis

5. 59 per cent of childbirth deliveries are not conducted in hospitals and 51 per cent are not safely done

6. 68.6 per cent of trafficked women and children are from rural India

7. Rural population's lack of online trading skills and poor technology infrastructure prevents them from taking advantage of e-commerce and m-commerce to market locally-produced goods such as handicrafts. According to eBay India, of the 4306 e-commerce hubs in the country, only 1015 hubs are rural hubs (<http://shopping.ebay.in/census/index.php>)

These problems can be addressed through large-scale deployment of connected services such as telemedicine, e-agriculture, online learning, mobile banking, e-commerce and e-governance, all of which pre-suppose the availability of affordable broadband connectivity.

Existing telecom connectivity scenario in rural India

The rural telecom connectivity scenario can be looked at in terms of its subscriber base, infrastructure and Internet usage.

Subscriber base. According to Department of Telecommunications (DoT) memo, as of August 2016, the country had 1053.86 million telephone connections of which there were 447.36 million (42 per cent) connections in rural areas and 606.50 million (58 per cent) in urban regions. Overall tele-density (number of telephones per 100 people) stands at 82.57 per cent with rural tele-density at 51 per cent and urban at 152 per cent.

Of the existing telephone connections, 97.7 per cent (1029.35 million) were wireless and only 2.3 per cent (24.51 million) were wireline. The corresponding wireline figure in August 2015 was 26.01 million, showing a yearly decline of 5.77 per cent.

In March 2016, the number of wireline subscribers was 25.22 million, of which, 4.32 million (17.13 per cent) were rural subscribers. The number of wireless subscribers was 1033.63 million, of which 444.84 million (43 per cent) were rural subscribers. Although the wireless percentages do not show much urban-rural disparity, the rural tele-density figures (51 per cent)